SENTIMENT ANALYSIS CHALLENGE

Sentiment Analysis Model Comparison on TripAdvisor Hotel Reviews

TextBlob • TextBlob + Naive Bayes • VADER • Flair

TripAdvisor Hotel Reviews dataset - Kaggle (approx. 20K reviews)

TripAdvisor Hotel Reviews

Source

tripadvisor_hotel_reviews.csv — free-text guest reviews paired with a 1–5 star rating.

Sampling

Random sample of 500 reviews (random_state=42) for a fast, reproducible benchmark.

Labelling — turning it binary

- Neutral reviews (rating = 3) removed for a clean Positive vs Negative task.
- rating ≥ 2 → 1 (positive)
- rating < 2 → 0 (negative)

500
reviews sampled

~91%
positive class

~9%
negative class

2
classes (binary)

⚠ **Strong class imbalance (~91% positive).** A naive "always positive" model already scores ~91% accuracy — so we judge models on macro F1.

Four Approaches, One Task

01

TextBlob

LEXICAL

Sum of word-level polarities (−1…+1). Simple, fast, no training.

02

TextBlob + Naive Bayes

PROBABILISTIC

Pre-trained NB classifier. Heavy inference — evaluated on 100 rows only.

03

VADER

RULE-BASED LEXICAL

Tuned lexicon + intensity/negation rules. Built for social text.

04

Flair

DEEP LEARNING

Pre-trained LSTM classifier. Context-aware, GPU-friendly, heavier.

Scoreboard: Accuracy · Precision · Recall · F1 · Speed

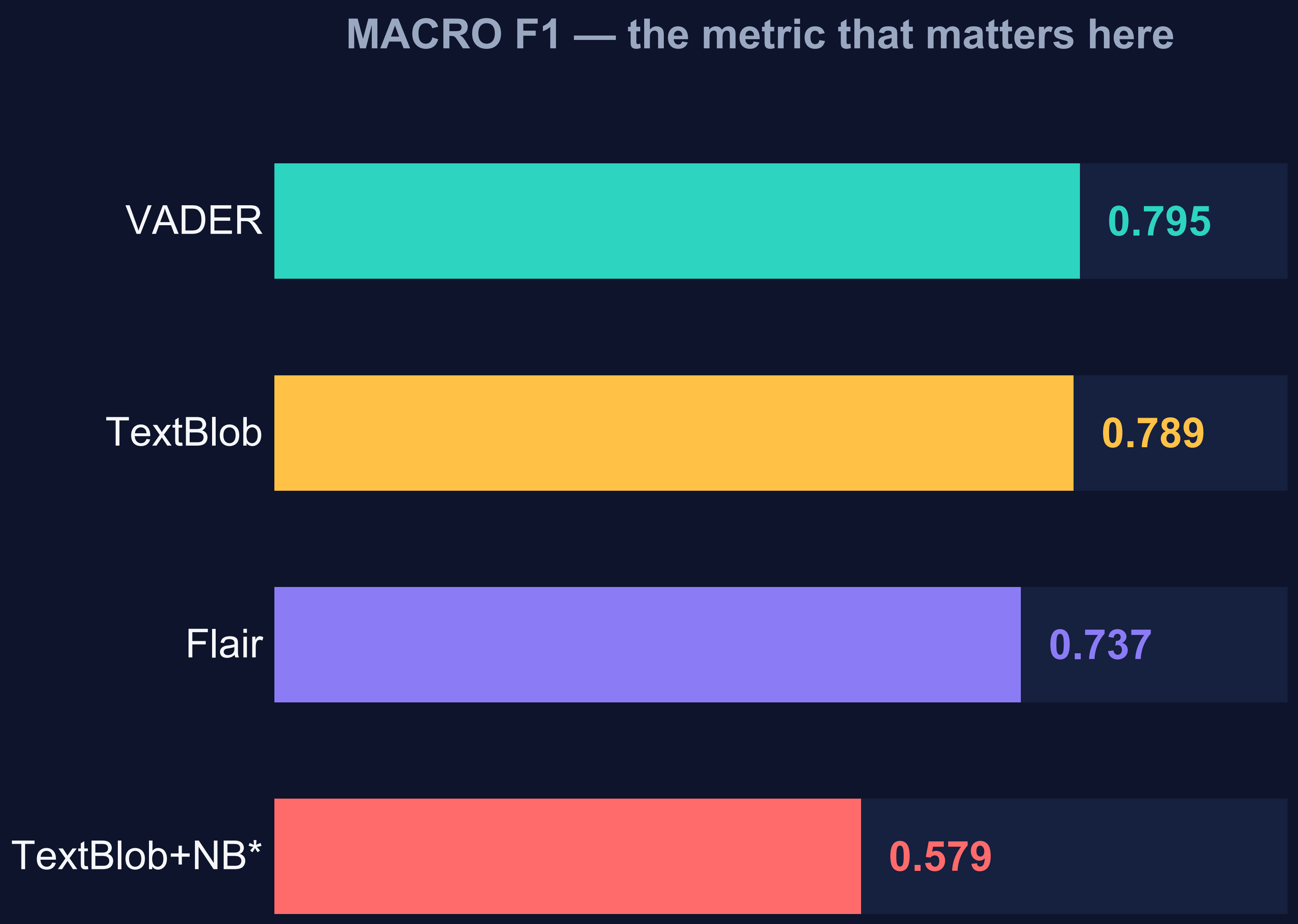
Model	Accuracy	Precision	Recall	F1 (macro)	Speed
TextBlob	0.939	0.844	0.752	0.789	Fast
VADER	0.931	0.789	0.801	**0.795**	**Fastest**
Flair	0.857	0.694	0.921	0.737	Slow
TextBlob + Naive Bayes*	0.920	0.636	0.561	0.579	Very slow

* Evaluated on 100 rows only (inference cost prohibitive) — not directly comparable. Speed is qualitative: rule-based (VADER/TextBlob) run in microseconds; Flair's LSTM is ~100–1000× slower.

Read past accuracy → on macro F1, VADER leads (0.795). TextBlob = most precise, Flair = most sensitive (recall 0.92) but least precise.

03 — RESULTS COMPARISON

What the Numbers Say



TextBlob — the cautious one

High precision (0.84), lower recall (0.75): right when it flags negative, but misses many true negatives.

VADER — the best balance

Highest macro F1 (0.795), precision ≈ recall. Lightweight, deterministic, no GPU.

Flair — the sensitive one

Recall 0.92 (catches almost every negative) but precision 0.69 → many false alarms, and far slower.

Winner: VADER

Best macro F1 (0.795) — and, rarely, also the cheapest to run.

Best balanced score

Top macro F1, with precision ≈ recall — strongest all-round profile on an imbalanced set.

Fits the data

Hotel reviews lean on explicit opinion words ("great", "terrible", "clean") — exactly VADER's strength.

Cheap at scale

Pure rules → microsecond, deterministic inference, no GPU. Trivial to scale to 10k req/s.

Strong, cheap, well-suited — but not flawless. Where does it still fall short? →

05 — LIMITATIONS

Where VADER Falls Short

Its 0.795 macro F1 hides real weaknesses, mostly on the minority negative class.

Fixed decision threshold

compound ≥ 0 = positive is arbitrary. On a ~91% positive set, this default leaves negative recall on the table.

Blind to context & sarcasm

Lexicon + local rules can't read irony or indirect negation ("couldn't have asked for better", "not bad at all").

Weak on the minority class

Negatives (~9%) drive most errors; macro F1 is pulled down by missed or mislabelled negative reviews.

No domain adaptation

A general social-media lexicon — hotel-specific terms ("dated", "noisy", "compact room") aren't weighted for this domain.

How can we push it further ?

Three steps, from quick win to richer model — keeping VADER's speed as the baseline.

01

Tune the threshold

Search the compound cut-off that maximises macro F1 instead of fixing it at 0. Zero added cost, immediate recall gain.

QUICK WIN

02

Hybrid model

Train a light logistic regression on VADER scores + TF-IDF features to learn the optimal boundary on our imbalanced data.

MEDIUM EFFORT

03

Fine-tuned transformer

Benchmark a domain-tuned BERT on the full dataset for context & sarcasm — the accuracy ceiling, if latency budget allows.

HIGH EFFORT

To summarise

On hotel reviews, VADER offers the best tradeoff and the best precision/recall balance at near-zero cost — TextBlob is more conservative, Flair the most sensitive but the most expensive.